Quiz-01: Engineering Physics (NPHI101), Monsoon 2025-26
Duration: 15 min * Total Marks: 10

Tick the correct answer | 1 mark each | Symbols have their usual meaning | Use the backside for rough work

Name: **Admission No:**

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1. The potential energy for a conservative force can be arbitrary upto the addition of a ___________ .
   (a) Constant number (b) Constant vector (c) Scalar function of the coordinates (d) None
2. The dimension of a Lagrangian is:
   (a) [Energy][time] (b) [time] (c) [length] (d) None
3. The walls of a glass container, constituting gas molecules, impose a constraint of type:
   (a) Holonomic (b) Non-holonomic (c) Rheonomous (d) None
4. Consider an optical cavity of length 1 m. What is the axial mode spacing?
   (a) 9.42×10^{8} rad/s (b) 9.42×10^{10} rad/s (c) 9.42×10^{8} Hz (d) 18.84×10^{8} rad/s
5. Consider an atomic radiative transition from $E_2 = 2.55$ eV to $E_2 = 1.00$ eV. What is the emitted wavelength?
   (a) 80 μm (b) 0.4 μm (c) 0.8 μm (d) None
6. Spontaneous emission is faster than stimulated emission process.
   (a) True (b) False (c) Can't be said
7. The life-time associated with the spontaneous decays from E_3 to E_2 and from E_3 to E_1 are 50 ns and 200 ns respectively. What is the life-time of E_2 for over all spontaneous decay from that level?
   (a) 250 ns (b) 40 ns (c) 25 ns (d) None

   (those who attempted and ticked any option will get marks)
8. Isothermal compressibility (κ) is
   (a) $\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_S$ (b) $\left(\frac{\partial V}{\partial P}\right)_T$ (c) $-\frac{1}{V}\left(\frac{\partial V}{\partial P}\right)_T$ (d) $-\frac{1}{T}\left(\frac{\partial V}{\partial P}\right)_T$
9. If *H(S, P)* is enthalpy of a system, then $\left(\frac{\partial V}{\partial S}\right)_P = \ldots\ldots\ldots$
   (a) $-\left(\frac{\partial V}{\partial P}\right)_T$ (b) $\left(\frac{\partial V}{\partial P}\right)_T$ (c) $\left(\frac{\partial S}{\partial P}\right)_T$ (d) $\left(\frac{\partial T}{\partial P}\right)_S$
10. The Gibbs free energy, G = U + PV + (..................)
    (a) F (b) TS (c) - TS (d) H